9. DIAGNOSTICS

9.1 INTRODUCTION TO DIAGNOSTICS

This section includes details of the different diagnostic methods provided within the software for a drive.

9.2 LED FAULT INDICATORS

Illumination of the WARNING or flashing TRIPPED indicator and extinguishing of the HEALTHY/STANDBY indicator indicate a fault condition.

The four LED indicators on the Keypad are repeated on the Keypad Harbour and give a first indication of drive status. The Drive and Keypad indicators are shown at Section 2.

9.3 WARNINGS

If the WARNING indicator is lit a problem has occurred which is not sufficiently serious to trip the drive. A warning code is stored in one of 10 locations in the Warning Record, parameters P10.00 to P10.09, the code stored in P10.00 being for the most recent warning. It is most important that all Warning codes are viewed to get a complete picture.

A full list of codes for WARNINGS is shown in Table 9-2.

NOTE: **Warnings are not latched and if the warning condition ceases, the WARNING indicator will extinguish. (At default configuration, Warning 1 is located at P1.06).**

9.4 TRIPS

If the TRIPPED indicator is flashing, a serious fault has occurred which has caused the drive to shut down. The drive may be faulty, but usually the drive has tripped in response to a problem on the plant, and the drive has protected the installation. Each time a trip occurs a Fault is stored in one of ten locations in the Active Trip record, parameters P10.10 to P10.19, the fault stored in P10.10 being for the most recent trip. It is most important that all Trip codes are viewed to get a complete picture.

A full list of codes for TRIPS is shown in Table 9-3.

NOTE: **Trips are latched and must be reset before the drive can be operated again.**

9.5 VIEWING WARNINGS AND TRIPS

Parameters in the drive report the trip or warning currently present, and other parameters hold a history of the last 10 trips. These parameters display codes and text that describe particular warnings or trips; the Keypad automatically displays these text messages. Menu 10 is dedicated to trips and warnings, but at default Menu 1 also has some of these parameters collected together for easy access.

Available parameters in Menu 1	Available parameters in Menu 10
P1.06 = FIRST WARNING	P10.00 - P10.09 = WARNINGS 1 to 10
P1.07 - P1.09 = FIRST 3 TRIPS	P10.10 - P10.19 = CURRENT TRIPS 1 to 10
	P10.20 - P10.29 = TRIP HISTORY 1 to 10

Table 9-1. – Viewing Warnings & Trips

Viewing using navigation keys

Navigate to one of the above parameters, either a Trip or a Warning and note the trip code and the trip text message – remember to view all, not just the first one.

Viewing using "help" key

a) When the drive is showing either a Trip or Warning, press ?.

b) The Keypad will display a diagnostic menu. Choose the relevant option.

c) See Sections 9.5.1 and 9.5.2 for what to do if a Trip or Warning occurs.

9.5.1 Action in the Event of a Warning

a) Press ? and select "2" – Display Warnings.

b) P10.00 will be displayed – note the first Warning! This is the problem that is causing the warning indication.

c) In turn, Display P10.01 to P10.09 and note any additional warnings. Any warnings in these locations will be for secondary problems and will help with diagnosis.

d) Refer to Table 9-2 and check the meaning of each warning. Take corrective action as necessary.

9.5.2 Action in the Event of a Trip

a) Press ? and select "2" – Display Trips.

b) P10.10 will be displayed – note the most recent Trip. This is the problem that has caused the trip indication. (For the default configuration, Trips 1/2 are located at P1.07/P1.08).

c) In turn, Display P10.11 to P10.19 and record any additional trips that may be present.

d) Refer to Table 9-3 and check the meaning of each fault code. Take corrective action as necessary.

e) See Section 9.5.3 for resetting trips.

9.5.3 Resetting Trips

From the Digital Inputs

a) From Default, press and release the button wired to DIGIN 6.

NOTE: CF9 (the Reset flag) may have been re-programmed, but at Default it is connected to DIGIN 6.

From the Drive Data Manager™ (Keypad)

a) Press ? and select Option 3 (Attempt Trip Reset).

Table 9-2. – Warning Fault Codes

Fault Code	Warning Description	Possible reason why
100	Excess output current	Drive output current > drive continuous rating
101	Motor thermostat (MTRIP)	MTRIP input is low (P2.10 = 1)
102	Motor I^2T	Accumulated I^2T product of motor overload has reached Warning level (P2.09 = 2 or 3)
103	Motor PTC	PTC resistance increasing and approaching trip level
104	DB resistor	Resistor has 25% remaining power dissipation capability
105	Reference loss	Failure of selected drive reference
106	FBC1 loss	Cable fallen out. Not configured correctly, either on drive, or network master.
107	High temperature - see P45.22	A temperature high warning has occurred – *see* P45.22 for more details ∗
108	Low temperature - see P45.22	A temperature low warning has occurred – *see* P45.22 for more details ∗
110	Backup reference loss	Failure of backup reference source

Table 9-2. – Warning Fault Codes

Fault Code	Warning Description	Possible reason why
111	CAN2 Warnings	Refer to MV3000 Second CAN Port technical manual, T1968
112	RS232 loss	RS232 serial link has timed out (P32.15 = 1)
113	RS485 loss	RS485 serial link has timed out (P32.55 = 1)
114	Over speed	Motor operating above over speed level (P29.00/P29.01)
115	Encoder loss	Signal from encoder suggests fault
116	Fieldbus Loss	The Fieldbus communication link has failed
117 to 119	CAN2 Warnings	Refer to MV3000 second CAN Port technical manual, T1968
120 to 125	Internal software fault	Fault detected in operating software
127	Back-up parameter set in use	If main parameter set is corrupt at power up, the drive reverts to the back-up parameter set (if stored via P99.16). SF11 is set High, and warning 127 is active, to indicate Back-up set is being used.
128	Load fault – High	Fault due to excessive load
129	Load fault – Low	Fault due to inadequate load
130 to 132 & 135	CAN Warnings	Refer to MV3000 CANopen Fieldbus Facility Technical Manual T2013 – see 1.6.
133	SFE: Mains frequency outside limits	Set by P52.09 and P52.10
134	SFE: Synch Loss (P99.01?)	It has not been possible to synchronise the mains e.g. because of an auxiliary supply fault. If attempt made to run with this Warning displayed the SFE will trip on Trip Code 94.
136	SFE: LCN feedback missing/CF25 not been enabled	The unit will therefore not respond to a run request.
137	SFE: Choke ptc Warning	Resistance is greater than the threshold set in P52.19 and P52.20 is set to the value 1.
138	DB inactive (P99.01?)	P23.01 set to 1 (Proportional) P99.01 not set to 0 (No Motor Control)
139	U-phase sharing failure	The U-phases between all of the connected DELTA units are not sharing current equally. Faulty wiring. Unequal wiring impedances/lengths. Loose wiring terminals Faulty DELTA unit.
140	V phase sharing failure	As for 139 except for V phase
141	W phase sharing failure	As for 139, except for W phase.
142	Mains Amplitude Warning	SFE Mode: The measured mains voltage amplitude is outside of the limits set in P52.22 and P52.23 Motor Control Mode: The measured mains voltage amplitude is such that AC Loss Ridethrough has been invoked.
143	Ethernet Channel 1 Loss	Ethernet Channel 1 has not received the appropriate protocol message within the time-out specified, see MV3000e Ethernet Interface Technical Manual T2034 for full details
144	Ethernet Channel 2 Loss	As for 143 except for Ethernet Channel 2.
145 & 146	Reserved - Future Use	
147	Download in Progress	A parameter set is being downloaded to the drive

NOTE: **Refer to Section 6.9 for details of the drive hardware.**

9.5.4 Trip Fault Codes

These are shown in Table 9-3. There are four classes of trip:

A = Auto-resettable trip (see Menu 28).
R = Manually resettable trip (see Section 9.5.3).
S = System trip.
N = Non-resettable trip.

Table 9-3. – Trip Fault Codes

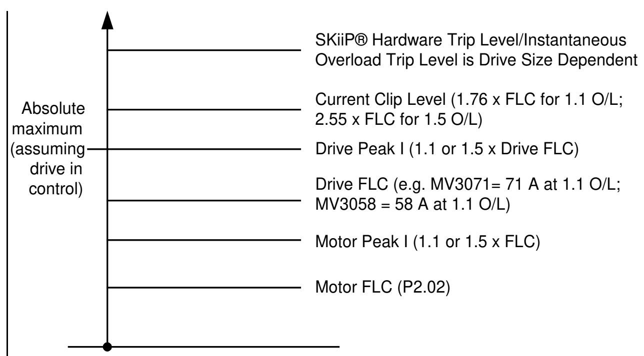
Fault Code	Name	Class	Description	Parameter No. to enable Auto-reset	What can cause a trip	Remarks
1	Interlock	A,R	Interlock Input at TB3/9 is low or open	P28.11	24 V not present on TB3/9. This is usually due to a plant E-stop or interlock being open	Check for open E-stops or interlocks outside the drive cubicle. Check that TB3 is not loose.
2	Reference Loss	A,R	No reference source can be found (either primary or backup)	P28.12	The drive receives references from analogue inputs, 4-20mA etc or Fieldbus link and the active reference has gone	Check more trip codes on Menu 10 by scrolling through. Extra trip codes will give clues to which reference has been lost.
3	DC Overvolts	A,R	DC Link Voltage exceeds the over voltage level: 780 V on a 380V-480VAC system 875V on a 525VAC system 1165V on a 690VAC system	P28.09	The DC Link volts monitored on P11.03 has risen; usually caused by the motor decelerating quickly and regenerating back into the drive. Check decelerating ramp rates are not too high or that any dynamic brake fitted is operating	If the decelerating ramp rate that stops the drive from tripping is too long, fit a DB. This is not a drive fault but an application error. Refer to Over/Under volts trip table in the appropriate product technical manual.
4	DC Undervolts	A,R	DC Link Voltage is less than the under voltage level: 400V on a 380V-480VAC system 450V on a 525VAC system 560V on a 690VAC system	P28.08		When a MicroCubicle™ CDC or a DELTA CDC is operating from 24 V auxiliaries the mains power is off and so the DC Link is not established.
5	Timed Over-current	A,R	Total I.T limit of the drive overload has been exceeded	P28.07		
6	Over-temperature -see P45.22 (pointed at by P10.10)	A,R	An Over-temperature trip has occurred – see P45.22 for more details	P28.13	Over temperature trips are generated by software in the CDC, by comparing temperature feedback to trip levels, or in the SKiiP® circuits in hardware, in case the software fails: - the temperature feedback is high - check Menu 45; - the drive ventilation is inhibited - check temperature, check air throughput against table in specification section of the appropriate product manual; - for liquid cooled products check that the coolant flow rates and volumes meet the specification in the product manual.	Refer to over temperature trips at Table 6-11. Menu 45 is dedicated to temperature feedback - some of the temperatures are also included at Menu 11. When a MicroCubicle is operating from 28V auxiliaries the SKiiP circuits are not powered up, which is where the trip originates, and the drive will show this trip.
7	Instantaneous Over-current	A,R	The drive output current has exceeded its maximum safe level – see diagram below for an indication of various trip levels. Note: When a Fault Code 7 occurs it is also likely that there will be a fault on any 8 to 13 U, V, W trip codes.	P28.06	The current feedback has exceeded the trip levels within the product: - motor faulty; - motor cable damage; - shock loads to system - however, the drive's internal current clipping ride-through function should stop this; - possible faulty transistors: - remove the motor cables, if the drive still trips then it is at fault internally; - for a DELTA system try exchanging the ribbons in the CDC and see if the trip 'moves'. If it does, the DELTA now reporting the trip is faulty. If it doesn't the CDC is faulty.	Refer to diagram under 'Description' for the various trip levels. When operating from 28V the drive will trip because the SKiiP circuits, within the product, are not powered. The exact list of trips will depend on the size of the drive powered by 28V.
				-		When operating from 24 V the drive will trip because the SKiiP® circuits, within the product, are not powered. The exact list of trips will depend on the size of the drive powered by 24 V.
8	Instantaneous Over-current on U-phase	A,R	U-phase transistor turned off by self-protection*	P28.06		
9	Hardware Over-temperature on U-phase	A,R	Over-temperature detected by U-phase output*	P28.13		

Table 9-3. – Trip Fault Codes

Fault Code	Name	Class	Description	Parameter No. to enable Auto-reset	What can cause a trip	Remarks
10	Instantaneous Over-current on V-phase	A,R	V-phase transistor turned off by self-protection*	P28.06		
11	Hardware Over-temperature on V-phase	A,R	Over-temperature detected by V-phase output*	P28.13		
12	Instantaneous Over-current on W-phase	A,R	W-phase transistor turned off by self-protection*	P28.06		
13	Hardware Over-temperature on W-phase	A,R	Over-temperature detected by W-phase output*	P28.13	The Over temperature Trips in Code 6 should react first. However, if the CDC is not functioning then the temperature may keep rising. The SKiiP® circuits have a hardwire temperature monitor for safety and these may trigger:- is the drive really hot - check Menus 11 and 45.- if YES check fans and ventilation requirements;- if NO then the drive may be at fault - e.g. change ribbon cables on DELTAs.	When operating from 24 V the drive will trip because the SKiiP® circuits, within the product, are not powered. The exact list of trips will depend on the size of the drive powered by 24 V.
14	Encoder Power Supply Fail	R	Encoder supply is internally supplied at 8 V then regulated down to the value set in P13.06 and 'I' limited, then if current flowing to the encoder is too high, trip here.	-	Check encoder cabling for damage – remove encoder connection at TB5 and try a reset to see if drive or cable is at fault – is encoder correct type? Refer to 'Encoder Selection' in appropriate product manual.	
15	Controller 15 V Fail	R	Internal supply voltage on the Control Board has failed. Failure of 15 V supply could result in: (a) no +15V for SKiiP which will result in any number of trips on the SKiiP module e.g. over temperature, over current.	-	Faulty CDC	
16	24 V Output Fail	R	The User +24 V output from the Control Electronics has failed, or been overloaded.	-	Disconnect User I/O terminals from the drive. See if 24 V recovers – check wiring if it does.	
17	Unidentified Power Interface Board	N.S	The drive cannot recognise the Power Interface (PIB) PCB, and therefore cannot run. It looks for an E²PROM on the PIB.	-	E²PROM failure on a MicroCubicle™ - the PIB is part of the CDC; On a DELTA system the PIB is within the controller's metal case.	
18	History Restore Fail	S	The Non-volatile History record cannot be recovered at power-up.	-	Can happen after a firmware upgrade and is OK. Is not critical if history record is not needed and therefore it can be ignored.	
19	New PCB	S	One of the printed circuit boards (pcb) in the drive has changed from when the drive was last switched off. Refer to Spares Manual supplied with new pcb.	-	Exchanging pcbs between drives Changing DELTA1; Changing DELTA pcbs. It is important that the correct spares procedures are followed.	
20	Parameter Edits Lost	S	The Non-volatile Parameter Edits cannot be recovered at power-up. All parameters have returned to their factory default values.	-	This trip is rare and is a warning. The DC Link on the drive may have been dragged down very quickly, preventing enough time for parameter storage on Power Off. Check for excessive DC Link Loads e.g. faulty DBs or DC Link schemes faulty.	
21	Motor Thermostat (MTRIP)	A,R	Motor thermostat has tripped. This trip can only occur if the thermostat is connected via a digital input to CF113 and P2.10 = 2. See Control Block Diagram Sheet 7	P28.10	CF113 is healthy when High, so n/c thermostats need to be inverted by placing a –ve sign in the edit.	Check motor fans and thermostat functionality.
22	Motor I²T	A,R	The motor I²T limit has been exceeded. This trip can only occur if P2.09 = 2.	P28.10	Is the motor free to rotate? Is the motor sized correctly for the load? Is P2.07 set correctly? Is P2.08 set correctly?	
23	RS232 Loss	A,R	RS232 serial link has timed out. This trip can only occur if P32.15 = 2.	P28.14	Check RS232 connections and wiring. Is the serial link manager plc OK? Check that the time-out period is set correctly.	
24	RS485 Loss	A,R	RS485 serial link has timed out. This trip can only occur if P32.55 = 2.	P28.14	Check RS485 connections and wiring. Is the serial link manager plc OK? Check that the time-out period is set correctly. Is P2.08 set correctly?	
25	Internal Reference Fail	R	Internal reference voltage has failed, CDC faulty	-	The CDC is faulty – replace.	
26	Under-temperature - see P45.22	A,R	An under-temperature trip has occurred – see P45.22 for more details	P28.13	Is the temperature actually low? *See* Menu 11 and 45 . If it is the drive cannot operate <0 °C. Temperature feedback comes from the SKiiP® Module. Check connections to the CDC.	
27	Keypad Loss/Removed	R	Keypad Communications have illegally halted. See Keypad removal at 2.2.5.	-	The Keypad has been removed when it was either in control (start/stop) or had reference control (Keypad ref).	

Table 9-3. – Trip Fault Codes

Fault Code	Name	Class	Description	Parameter No. to enable Auto-reset	What can cause a trip	Remarks
28	Current Imbalance	R	The 3 output phases are imbalanced in current This trip can only occur in VVVF mode (P99.01 = 1).	-	If the current is different by 20% in any 1 phase the drive trips. Disconnect motor cables and see if the trip remains - if YES then = drive; - if NO then = motor; Check motor. Check motor cables.	Check fans and ventilation. Check motor loads and cables - can disconnect motor cables and run drives to see if trip goes away - if it does the drive is not at fault.
29	Precharge failure	A,R	The Power Interface Board received no pre-charge contactor close signal. On a DELTA, check the rectifier wiring.	P28.08	24 V Precharge Acknowledge on PL12/9 has not been detected – wiring error or relay on DELTA Rectifier faulty or 24 V is down.	
30	Drive ID Violation	R	P99.13 is not equal to P99.14	-	Simple editing error. FIP or Fieldbus parameter set sent to wrong drive.	
31	DELTA 1 Over-current on U-phase	A,R	U-phase transistor on DELTA 1 turned off by self-protection.	P28.06	The Over current and Over temperature trips are operated by the SKiiP® Modules. Either the temperature or current is high (check Menus 11 and 45) or the SKiiP® Modules are faulty (move CDC ribbons for the DELTAs to see if the Trips move) or the controller is faulty. If the trips do not move with the DELTA ribbons and not the motor then it must be the controller. Checking if drive is faulty: 1. Remove motor cables from drive end; 2. 'Run' the drive in VVVF Mode and see if trips occur - if not drive is OK - if it does it may be controller or DELTA at fault. Checking if DELTA is faulty: 1. With the power OFF, move DELTA ribbons in the DELTA CDC; 2. Power ON and run the drive; 3. If the drive trips again and the trip has moved with the DELTA then the DELTA is at fault; If the drive trips on the same trip then the DELTA is OK but the CDC is at fault.	
32	DELTA 1 Over-temp on U-phase	A,R	Over-temperature detected on DELTA 1 U-phase.*	P28.13	See Fault Code 31 above.	
33	DELTA 1 Over-current on V-phase	A,R	V-phase transistor on DELTA 1 turned off by self-protection.	P28.06	See Fault Code 31 above.	
34	DELTA 1 Over-temp on V-phase	A,R	Over-temperature detected on DELTA 1 V-phase.*	P28.13	See Fault Code 31 above.	
35	DELTA 1 Over-current on W-phase	A,R	W-phase transistor on DELTA 1 turned off by self-protection	P28.06	See Fault Code 31 above.	
36	DELTA 1 Over-temp on W-phase	A,R	Over-temperature detected on DELTA 1 W-phase.*	P28.13	See Fault Code 31 above.	
37	DELTA 1 External trip	R	External trip detected on DELTA 1, TB1.	-	Only possible on Liquid Cooled DELTA (LCD) – this TB is used to detect fan failure on the LCD. See DC Link Overvoltage Trip 3. Could be faulty SMPS - exchange with another.	
38	DELTA 1 Over-voltage	A,R	DC Link over voltage detected on DELTA 1 – signal from the SMPS	P28.09		
39	DELTA 1 Enhanced DIB Trip	A,R	Reported if an enhanced trip from new Delta Interface Board (20X4381) is detected by an older controller.	P28.06	Controllers made after September 2005 show trip codes 240 to 249 instead.	
40	DELTA 2 Over-current on U-phase	A,R	U-phase transistor on DELTA 2 turned off by self-protection.	P28.06	Checks for Overcurrent on DELTA 2 are the same as those for DELTA 1 – refer to details at Fault Code 31.	
41	DELTA 2 Over-temp on U-phase	A,R	Over-temperature detected on DELTA 2 U-phase.*	P28.13	Checks for Overtemperature on DELTA 2 are the same as those for DELTA 1 – refer to details at Fault Code 31.	
42	DELTA 2 Over-current on V-phase	A,R	V-phase transistor on DELTA 2 turned off by self-protection.	P28.06	See Fault Code 40 above.	
43	DELTA 2 Over-temp on V-phase	A,R	Over-temperature detected on DELTA 2 V-phase.*	P28.13	See Fault Code 40 above.	
44	DELTA 2 Over-current on W-phase	A,R	W-phase transistor on DELTA 2 turned off by self-protection	P28.06	see Fault Code 40 above.	
45	DELTA 2 Over-temp on W-phase	A,R	Over-temperature detected on DELTA 2 W-phase.*	P28.13	see Fault Code 40 above.	
46	DELTA 2 External trip	R	External trip detected on DELTA 2, TB1.	-	See details for Fault Code 37.	
47	DELTA 2 Over-voltage	A,R	DC Link over voltage detected on DELTA 2.	P28.09		

Table 9-3. – Trip Fault Codes

Fault Code	Name	Class	Description	Parameter No. to enable Auto-reset	What can cause a trip	Remarks
48	DELTA 2 Over-current	A,R	Reported if an enhanced trip from new Delta Interface Board (20X4381) is detected by an older controller.	P28.06	Controllers made after September 2005 show trip codes 250 to 259 instead.	
49	DELTA 3 Over-current on U phase	A,R	U-phase transistor on DELTA 3 turned off by self-protection	P28.06	Checks for Overcurrent on DELTA 3 are the same as those for DELTA 1 - refer to details at Fault Code 31.	
50	DELTA 3 Over-temp on U-phase	A,R	Over-temperature detected on DELTA 3 U-phase.*	P28.13	Checks for Over temperature on DELTA 3 are the same as those for DELTA 1 - refer to details at Fault Code 31.	
51	DELTA 3 Over-current on V-phase	A,R	V-phase transistor on DELTA 3 turned off by self-protection.	P28.06	See Fault Code 49 above.	
52	DELTA 3 Over-temp on V-phase	A,R	Over-temperature detected on DELTA 3 V-phase.*	P28.13	See Fault Code 49 above.	
53	DELTA 3 Over-current on W-phase	A,R	W-phase transistor on DELTA 3 turned off by self-protection	P28.06	See Fault Code 49 above.	
54	DELTA 3 Over-temp on W-phase	A,R	Over-temperature detected on DELTA 3 W-phase.*	P28.13	See Fault Code 49 above.	
55	DELTA 3 External trip	R	External trip detected on DELTA 3, TB1.	-	See details for Fault Code 37.	
56	DELTA 3 Over-voltage	A,R	DC Link over voltage detected on DELTA 3.	P28.09		
57	Overspeed	A,R	The motor is operating above its set over speed level. This trip can only occur if P29.02 = 3.	P28.16	Speed feedback level in P9.01 is > over speed level in P29.00, P29.01. Is the encoder OK? Is the motor being overhauled? If in torque control, the motor may be running off due to mechanical failure of the load.	
58	Current Control Fault	R	Drive cannot control the motor current as required. Check the motor model, mains voltage, encoder connections and check for earth faults.	-	Value set in P12.02 is too high, or mains supply voltage is lower than thought. Poor motor model - do a CAL run. Encoder connections - mechanical and electrical are poor. Ensure the motor is off-load - the drive checks this at the beginning of the test. Motor parameters have been entered manually and are incorrect - do a CAL run.	Are the PID settings for Speed Amplifier OK? It may be unstable.
59	Motor Calibration Failure	R	Motor Calibration Test (P3.12 or 12.03 = 3) failed. Check the motor and encoder connections.	-		Motor must be off load.
60	Unsuitable Motor	R	The Motor Parameters entered do not describe a motor within the control capabilities of the drive.	-	The drive checks for three conditions to flag encoder loss: Maximum speed change (set in P13.02); Maximum reversals (set in P13.19); Reversible threshold (set in P13.20). Check encoder electrical and mechanical connections - ensure encoder cable has a continuous screen and is earthed at both ends.	
61	Encoder Loss	R	Encoder Signal has been lost, or cannot be relied upon. This trip can only occur if P13.02 = 3. See Control Block Diagram, Sheet 6.	-	The drive checks for three conditions to flag encoder loss: Maximum speed change (set in P13.02); Maximum reversals (set in P13.19); Reversible threshold (set in P13.20). Check encoder electrical and mechanical connections - ensure encoder cable has a continuous screen and is earthed at both ends. CF10 has been connected to something by the user – ensure that whatever it is connected to is healthy (i.e. producing a LO on CF10).	
62	User Trip 1	A,R	Control Flag 10 = 1	P28.15		
63	Fieldbus Loss	R	The Fieldbus communication link has failed.	-	FIP, PROFIBUS, are Fieldbus communication links – the one in use has failed.	
64	Load Fault - High	A,R	The drive has detected a load fault due to excessive load.	P28.17	Menu 43 allows the drive to monitor the load – check mechanics for bearing failure or coupling breakages etc.	
65	Load Fault - Low	A,R	The drive has detected a load fault due to inadequate load.	P28.17		
66	Motor PTC trip	R	Excessive temperature detected by PTC on TB5, pin 1. See Menu 2.	P28.10	Is the motor stalled? Is the motor fan faulty? Is the motor PTC faulty? Is the PTC wiring and connectors OK?	
67	DB resistor trip	A,R	Drive estimates the DB resistor is too hot.	P28.18	Only possible for a MV DB. Drive bases estimation on information typed into P23.01 to P23.03.	
68	Unknown Drive Size	N.S	The software was unable to read the E^2 PROM on one, or more, of the pcbs.	-	E^2 PROMs exist on the SMPS, the DELTA PIBs and the DELTA DIBs.	

Table 9-3. – Trip Fault Codes

Fault Code	Name	Class	Description	Parameter No. to enable Auto-reset	What can cause a trip	Remarks
69	Configuration Trip	N.S	The drive is not correctly configured to operate in a DELTA system. DELTAs are not connected to the DELTA Controller correctly.	-	The DELTAs must be plugged in from DELTA 1 slot to DELTA 6 with no gaps; The DELTAs must all be the same type; The SMPS units on the DELTAs must all be the same type.	
70	Datumize Error	R	Datumizing (Position Control Only) encountered limit switches in the wrong order. See Position Control.	-	Ensure that the HI and LO position limit switches are wired and programmed correctly to CF87 and CF86. Datumizing speeds set in P36.35 to P36.37 are too high – see Datumizing 6.36.3.	
71	Speed Feedback Loss	R	Only applies to Speed Feedback Vector Control Mode (P99.01 = 2, and P13.00 = 1 & 2).	-	The selected speed feedback is no longer present, or backup has been selected, and it has not been programmed or connected.	
72	Over Frequency	R	Allowed output frequency has been exceeded. 250Hz at switching frequencies >= 2.5kHz 125Hz at switching frequency = 1.25kHz	-	Absolute limit to prevent IGBTs from being damaged.	
73	User Trip 2	A,R	Control Flag 112 = 1	P28.15	See User Trip 1 to Trip Code 62.	
74	DELTA 3 Over-current	A,R	Reported if an enhanced trip from new Delta Interface Board (20X4381) is detected by an older controller.	P28.06	Controllers made after September 2005 show trip codes 260 to 269 instead.	
75	DELTA 4 Over-current on U-phase	A,R	U-phase transistor on DELTA 4 turned off by self-protection	P28.06	Checks for Overcurrent on DELTA 4 are the same as those for DELTA 1 - refer to details at Fault Code 31.	
76	DELTA 4 Over-temp on U-phase	A,R	Over-temperature detected on DELTA 4 U-phase.*	P28.13	Checks for over temperature on DELTA 4 are the same as those for DELTA 1 - refer to details at Fault Code 31.	
77	DELTA 4 Over-current on V-phase	A,R	V-phase transistor on DELTA 4 turned off by self-protection.	P28.06	See Fault Code 75 above.	
78	DELTA 4 Over-temp on V-phase	A,R	Over-temperature detected on DELTA 4 V-phase.*	P28.13	See Fault Code 76 above.	
79	DELTA 4 Over-current on W-phase	A,R	W-phase transistor on DELTA 4 turned off by self-protection	P28.06	See Fault Code 75 above.	
80	DELTA 4 Over-temp on W-phase	A,R	Over-temperature detected on DELTA 4 W-phase.*	P28.13	see Fault Code 76 above.	
81	DELTA 4 External trip	R	External trip detected on DELTA 4, TB1.	-	See details for Fault Code 37.	
82	DELTA 4 Over-voltage	A,R	DC Link over voltage detected on DELTA 4.	P28.09		
83	DELTA 4 Over-current	A,R	Reported if an enhanced trip from new Delta Interface Board (20X4381) is detected by an older controller.	P28.06	Controllers made after September 2005 show trip codes 270 to 279 instead.	
84	DELTA 5 Over-current on U-phase	A,R	U-phase transistor on DELTA 5 turned off by self-protection	P28.06	Checks for Overcurrent on DELTA 5 are the same as those for DELTA 1 - refer to details at Fault Code 31.	
85	DELTA 5 Over-temp on U-phase	A,R	Over-temperature detected on DELTA 5 U-phase.*	P28.13	Checks for over temperature on DELTA 5 are the same as those for DELTA 1 - refer to details at Fault Code 31.	
86	DELTA 5 Over-current on V-phase	A,R	V-phase transistor on DELTA 5 turned off by self-protection.	P28.06	See Fault Code 84 above.	
87	DELTA 5 Over-temp on V-phase	A,R	Over-temperature detected on DELTA 5 V-phase.*	P28.13	See Fault Code 85 above.	
88	DELTA 5 Over-current on W-phase	A,R	W-phase transistor on DELTA 5 turned off by self-protection	P28.06	See Fault Code 84 above.	
89	DELTA 5 Over-temp on W-phase	A,R	Over-temperature detected on DELTA 5 W-phase*	P28.13	See Fault Code 85 above.	
90	DELTA 5 External trip	R	External trip detected on DELTA 5, TB1.	P25.19	See details for Fault Code 37.	
91	DELTA 5 Over-voltage	A,R	DC Link over voltage detected on DELTA 5.	P28.09		
92	DELTA 5 Over-current	A,R	Reported if an enhanced trip from new Delta Interface Board (20X4381) is detected by an older controller.	P28.06	Controllers made after September 2005 show trip codes 280 to 289 instead.	
93	SFE: Mains frequency detected outside of limits	A,R	Mains frequency has been detected outside of the limits set by P52.08 and P52.11	P28.19	The Mains Monitor Unit may not be connected correctly. The mains frequency may be varying.	
94	SFE: Synchronisation to mains has failed	A,R	Synchronisation to the mains has failed due to: a) permanent loss of mains signal, or b) temporary interruption of mains signal, or c) mains frequency has changed at a rate greater than 2%/s.	P28.19	The Mains Monitor Unit may not be connected at all, or connected correctly.	
95	SFE: Mains Voltage Monitor Loss	A,R	Mains Voltage Monitor Loss	P28.19	A ribbon cable may have become disconnected	
96	SFE: Auxiliary Phase Detection Loss	A,R	Auxiliary Phase Detection Loss	P28.19	There may be a missing phase on an Input Monitor or a mismatch of amplitude and phasing on the incoming phases.	
97	Pre-release Expiry	R	Pre-release Software time limit expired	-	A pre-release version of firmware has been left in the drive – update to latest version.	
98	SFE: Choke ptc trip	A,R	Choke PTC resistance is greater than the threshold set in P52.19 and P52.20 is set to the value 2.	P28.19	The SFE Choke is either very hot; Or its detector (PTC) has failed; Or its detector (PTC) is not wired correctly.	

Table 9-3. – Trip Fault Codes

Fault Code	Name	Class	Description	Parameter No. to enable Auto-reset	What can cause a trip	Remarks
99	Unknown trip			-		
100-	Reserved for Warning Codes			-	The controller has detected an error in the PWM Pattern. CDC is at fault. Check DB resistor wiring, or DB resistor, maybe DB Family. Check DB resistor wiring and resistor. The DB may be hot due to short circuit of cables or DB may be faulty. Checks for Overcurrent on DELTA 6 are the same as those for DELTA 1 - refer to details at Fault Code 31.	
149	See Warning Codes			-		
150	PWM error	N.S	Error detected in Drive's PWM. Report to **GE Power Conversion**	-	The controller has detected an error in the PWM Pattern. CDC is at fault.	
151	DB Trip	A,R	Over-current from DB unit – detected by the SKiiP® Modules.	P28.06	Check DB resistor wiring, or DB resistor, maybe DB Family.	
152	DB hardware Over-temp trip	A,R	Over-temperature from DB unit – detected by the SKiiP® Modules.	P28.13	Check DB resistor wiring and resistor. The DB may be hot due to short circuit of cables or DB may be faulty.	
153	DELTA 6 Over-current on U-phase	A,R	U-phase transistor on DELTA 6 turned off by self-protection	P28.06	Checks for Overcurrent on DELTA 6 are the same as those for DELTA 1 - refer to details at Fault Code 31.	
154	DELTA 6 Over-temp on U-phase	A,R	Over-temperature detected on DELTA 6 U-phase.*	P28.13	Checks for over temperature on DELTA 6 are the same as those for DELTA 1 - refer to details at Fault Code 31.	
155	DELTA 6 Over-current on V-phase	A,R	V-phase transistor on DELTA 6 turned off by self-protection.	P28.06	See Fault Code 153 above.	
156	DELTA 6 Over-temp on V-phase	A,R	Over-temperature detected on DELTA 6 V-phase.*	P28.13	See Fault Code 154 above.	
157	DELTA 6 Over-current on W-phase	A,R	W-phase transistor on DELTA 6 turned off by self-protection	P28.06	See Fault Code 153 above.	
158	DELTA 6 Over-temp on W-phase	A,R	Over-temperature detected on DELTA 6 W-phase.*	P28.13	See Fault Code 154 above.	
159	DELTA 6 External trip	R	External trip detected on DELTA 6, TB1.	-	See details for Fault Code 37.	
160	DELTA 6 Over-voltage	A,R	DC Link over voltage detected on DELTA 6.	P28.09		
161	DELTA 6 Over-current	A,R	Reported if an enhanced trip from new Delta Interface Board (20X4381) is detected by an older controller.	P28.06	Controllers made after September 2005 show trip codes 290 to 299 instead.	
162	DB 1 Over-current U	A,R	U-phase transistor on DB 1 turned off by self-protection.	P28.06	DELTA Bridges can be used as DBs by connecting them into the relevant connections or the CDC Controller; the trips are therefore generated in the same way as a normal DELTA, so refer to DELTA 1 Codes. However, over currents can be caused by faulty resistors, or resistor wiring, as there is no motor here – check wiring and resistors for damage. Move the ribbons from DB1 to DB2. See if the trip moves with the DB or the Controller.	
163	DB 1 Over-temp U	A,R	Over-temperature detected on DB 1 U-phase.	P28.13		
164	DB 1 Over-current V	A,R	V-phase transistor on DB 1 turned off by self-protection.	P28.06		
165	DB 1 Over-temp V	A,R	Over-temperature detected on DB 1 V-phase.	P28.13		
166	DB 1 Over-current W	A,R	W-phase transistor on DB 1 turned off by self-protection.	P28.06		
167	DB 1 Over-temp W	A,R	Over-temperature detected on DB 1 W-phase.	P28.13		
168	DB 1 External trip	R	External trip detected on DB 1, TB1.	-		
169	DB 1 Over-voltage	A,R	DC Link over voltage detected on DB 1.	P28.09		
170	DB 1 Over-current	A,R	Over-current detected on DB 1.	P28.06		
171	DB 2 Over-current U	A,R	U-phase transistor on DB 2 turned off by self-protection.	P28.06		
172	DB 2 Over-temp U	A,R	Over-temperature detected on DB 2 U-phase.	P28.13		
173	DB 2 Over-current V	A,R	V-phase transistor on DB 2 turned off by self-protection.	P28.06		
174	DB 2 Over-temp V	A,R	Over-temperature detected on DB 2 V-phase.	P28.13		
175	DB 2 Over-current W	A,R	W-phase transistor on DB 2 turned off by self-protection.	P28.06		
176	DB 2 Over-temp W	A,R	Over-temperature detected on DB 2 W-phase.	P28.13		
177	DB 2 External trip	R	External trip detected on DB 2, TB1.	-		
178	DB 2 Over-voltage	A,R	DC Link over voltage detected on DB 2.	P28.09		
179	DB 2 Over-current	A,R	Over-current detected on DB 2.	P28.06		
180	PIB FPGA config. Failure	N.S	Internal failure. Report to **GE Power Conversion**	-	Identity in E^2PROM on MicroCubicle™ SMPS is unrecognisable – faulty SMPS or corrupt E^2PROM or maybe loose ribbon connections between CDC and SMPS.	

Table 9-3. – Trip Fault Codes

Fault Code	Name	Class	Description	Parameter No. to enable Auto-reset	What can cause a trip	Remarks
181	Unidentified SMPS	N.S	System error as described. Check wiring.	-		
182	Unidentified DELTA 1	N.S	Identity information in DIB is unrecognisable.	-		
183	Unidentified SMPS 1	N.S	Identity information in DELTA SMPS is unrecognisable.	-		When a DELTA CDC is operating from 24 V auxiliaries the CDC is unable to read the E²PROM on the DELTA DIB (DELTA Interface Board) so this trip will occur. The trip can be masked by CF108.
184	Incorrect DELTA 1	N.S	The complete DELTA is incorrect type	-	This means that the SMPS and the DELTA are not compatible. There may be a high voltage SMPS with a low voltage DELTA or vice versa – check correct SMPS type for DELTA.	When a DELTA CDC is operating from 24 V auxiliaries the CDC is unable to read the E²PROM on the DELTA SMPS (Switched Mode Power Supply) so this trip will occur. The trip can be masked by CF108.
185	Unidentified DELTA 2	N.S	See 182.	-	See 182	When a DELTA CDC is operating from 24 V auxiliaries the CDC is unable to read the E²PROM on the DELTA DIB (DELTA Interface Board) so this trip will occur.
186	Unidentified SMPS 2	N.S	See 183.	-		When a DELTA CDC is operating from 24 V auxiliaries the CDC is unable to read the E²PROM on the DELTA SMPS (Switched Mode Power Supply).
187	Incorrect DELTA 2	N.S	See 184.	-		
188	Unidentified DELTA 3	N.S	See 182.	-		
189	Unidentified SMPS 3	N.S	See 183.	-		
190	Incorrect DELTA 3	N.S	See 184.	-		
191	Unidentified DELTA 4	N.S	See 182.	-		
192	Unidentified SMPS 4	N.S	See 183.	-		
193	Incorrect DELTA 4	N.S	See 184.	-		
194	Unidentified DELTA 5	N.S	See 182.	-		
195	Unidentified SMPS 5	N.S	See 183.	-		
196	Incorrect DELTA 5	N.S	See 184.	-		
197	Unidentified DELTA 6	N.S	See 182.	-		
198	Unidentified SMPS 6	N.S	See 183.	-		
199	Incorrect DELTA 6	N.S	See 184.	-		
200	CAN 1 Loss, see P61.43	R	CAN is not communicating correctly	-	Some possible causes of a trip are:CAN cable disconnected;No other active nodes on the CAN Network.Refer to T2013 CANopen Fieldbus Facility Technical Manual for more details.	
201	Unsuitable DB Load	N.R	Load too high (resistor value too low) on DB mode, on IGBT Bridge	-	When using a Power Bridge as a DB (P23.19 = 1) incorrect value of DB resistors will trigger this trip. Check resistor values; Check DB resistor wiring.	
202	DELTA Rectifier Thermostat Trip	A,R	Thermostat connections (PL12/3 and PL12/4) on DELTA Controller are open circuit.	P28.13	Missing or damaged wiring between DELTA Controller (PL12/3/4) and DELTA Rectifier (TB1/3/4).	
203	CAN 2 LOSS See P65.43	R	As Trip 200	-	As Trip 200	
204	Fieldbus Coupler 1 Loss	R	The Fieldbus communication associated with Menu 74 has failed	-	Cable fallen out Not configured correctly - either on drive or network master.	
205	Reserved			-		
206	Reserved			-		
207	U-Phase Sharing Failure	R	Imbalance between DELTA U-Phase Currents	-	The U- Phases between all of the connected DELTA units are not sharing current equally: Unequal wiring impedances/lengths Loose wiring terminals Faulty DELTA unit	
208	V-Phase Sharing Failure	R	Imbalance between DELTA V-Phase Currents	-	As Trip 207, except for V-Phase	
209	W-Phase Sharing Failure	R	Imbalance between DELTA W-Phase Currents	-	As Trip 207, except for W-Phase	
210	Mains Amplitude Trip	R	The mains supply amplitude has been outside the limits set by P52.22 and P52.23 for longer than the time set by P52.24	-		

Table 9-3. – Trip Fault Codes

Fault Code	Name	Class	Description	Parameter No. to enable Auto-reset	What can cause a trip	Remarks
211	Not 40MHz CDC	N.S	The firmware fitted is only compatible with a 40MHZ CDC and the CDC in use is an older 25MHz version	-		
212	Undiagnosed Trip DELTA 1	R	Controller has detected a trip from DELTA 1 but is unable to diagnose details of the trip.	-	High amounts of electrical noise. Poor earthing. Incorrect screening of ribbon cables.	
213	Undiagnosed Trip DELTA 2	R	See 212	-		Later Delta Interface Boards (20X4381 or 20X4384) should fix this problem.
214	Undiagnosed Trip DELTA 3	R	See 212	-		
215	Undiagnosed Trip DELTA 4	R	See 212	-		
216	Undiagnosed Trip DELTA 5	R	See 212	-		
217	Undiagnosed Trip DELTA 6	R	See 212	-		
218	Undiagnosed Trip DB1	R	See 212	-		
219	Undiagnosed Trip DB2	R	See 212	-		
220	Ethernet 1 Loss	R	Ethernet Channel 1 has not received the appropriate protocol message (see P86.34) within the timeout specified by P86.33	-		
221	Ethernet 2 Loss	R	Ethernet Channel 2 has not received the appropriate protocol message (see P86.84) within the timeout specified by P86.83	-		
222	Torque Prove Fail	R	Torque has not been proved before the Torque Prove timeout has elapsed	-		
223	Brake Release Fail	R	Brake has not been released before the Brake Release timer has expired	-		
224	Download in Progress	R	Running is prohibited as Drive Coach PC package has not terminated a parameter download	-	Incorrect termination of parameter download from Drive Coach	
225 - 228	Reserved			-		
229	DSP Comms Failure	R	Communications between CDC and DSP have failed.	-		
230	User Alert 1	R	Control Flag 198 = 1	-	CF198 has been connected to something by the user – ensure that whatever it is connected to is healthy (i.e. producing a LO on CF198).	
231	User Alert 2	R	Control Flag 199 = 1	-	CF199 has been connected to something by the user – ensure that whatever it is connected to is healthy (i.e. producing a LO on CF199).	
232	User Alert 3	R	Control Flag 200 = 1	-	CF200 has been connected to something by the user – ensure that whatever it is connected to is healthy (i.e. producing a LO on CF200).	
233	User Alert 4	R	Control Flag 201 = 1	-	CF201 has been connected to something by the user – ensure that whatever it is connected to is healthy (i.e. producing a LO on CF201).	
234	User Alert 5	R	Control Flag 202 = 1	-	CF202 has been connected to something by the user – ensure that whatever it is connected to is healthy (i.e. producing a LO on CF202).	
235	User Alert 6	R	Control Flag 203 = 1	-	CF203 has been connected to something by the user – ensure that whatever it is connected to is healthy (i.e. producing a LO on CF203).	
236	User Alert 7	R	Control Flag 204 = 1	-	CF204 has been connected to something by the user – ensure that whatever it is connected to is healthy (i.e. producing a LO on CF204).	
237	User Alert 8	R	Control Flag 205 = 1	-	CF205 has been connected to something by the user – ensure that whatever it is connected to is healthy (i.e. producing a LO on CF205).	
238	User Alert 9	R	Control Flag 206 = 1	-	CF206 has been connected to something by the user – ensure that whatever it is connected to is healthy (i.e. producing a LO on CF206).	
239	User Alert 10	R	Control Flag 207 = 1	-	CF207 has been connected to something by the user – ensure that whatever it is connected to is healthy (i.e. producing a LO on CF207).	
240	Delta 1 DIB 5v Fail	R	Internal power supply failure on Delta DIB.	-	Faulty SMPS. Fault DIB.	
241	Delta 1 DIB +15v Fail	R	Internal power supply failure on Delta DIB.	-	Faulty SMPS. Fault DIB.	
242	Delta 1 DIB -15v Fail	R	Internal power supply failure on Delta DIB.	-	Faulty SMPS. Fault DIB.	
243	Delta 1 DIB Fan Fail	R	Speed feedback failure from capacitor fan for this Delta.	-	Faulty, blocked or jammed capacitor fan. If no fan is present or required, is the jumper link, TP14, on the Delta DIB in the correct position to disable the fan trip detection?	
244	Delta 1 DIB Over Freq. U	R	Erroneous PWM pulses have been detected by the DIB, on this phase, resulting in PWM frequency too high.	-	Incorrect screening of ribbons between controller and DIB. Screen is not connected to metalwork.	
245	Delta 1 DIB Over Freq. V	R	As Trip 244.	-	As Trip 244.	

Table 9-3. – Trip Fault Codes

Fault Code	Name	Class	Description	Parameter No. to enable Auto-reset	What can cause a trip	Remarks
246	Delta 1 DIB Over Freq. W	R	As Trip 244.	-	As Trip 244.	
247	Delta 1 DIB Td Error U	R	Erroneous PWM pulses have been detected by the DIB, on this phase, resulting upper/lower dead-time violation commands.	-	Incorrect screening of ribbons between controller and DIB. Screen is not connected to metalwork.	
248	Delta 1 DIB Td Error V	R	As Trip 247.	-	As Trip 247.	
249	Delta 1 DIB Td Error W	R	As Trip 247.	-	As Trip 247.	
250	Delta 2 DIB 5v Fail	R	Internal power supply failure on Delta DIB.	-	Faulty SMPS. Fault DIB.	
251	Delta 2 DIB +15v Fail	R	Internal power supply failure on Delta DIB.	-	Faulty SMPS. Fault DIB.	
252	Delta 2 DIB -15v Fail	R	Internal power supply failure on Delta DIB.	-	Faulty SMPS. Fault DIB.	
253	Delta 2 DIB Fan Fail	R	Speed feedback failure from capacitor fan for this Delta.	-	Faulty, blocked or jammed capacitor fan. If no fan is present or required, is the jumper link, TP14, on the Delta DIB in the correct position to disable the fan trip detection?	
254	Delta 2 DIB Over Freq. U	R	Erroneous PWM pulses have been detected by the DIB, on this phase, resulting in PWM frequency too high.	-	Incorrect screening of ribbons between controller and DIB. Screen is not connected to metalwork.	
255	Delta 2 DIB Over Freq. V	R	As Trip 244.	-	As Trip 244.	
256	Delta 2 DIB Over Freq. W	R	As Trip 244.	-	As Trip 244.	
257	Delta 2 DIB Td Error U	R	Erroneous PWM pulses have been detected by the DIB, on this phase, resulting upper/lower dead-time violation commands.	-	Incorrect screening of ribbons between controller and DIB. Screen is not connected to metalwork.	
258	Delta 2 DIB Td Error V	R	As Trip 247.	-	As Trip 247.	
259	Delta 2 DIB Td Error W	R	As Trip 247.	-	As Trip 247.	
260	Delta 3 DIB 5v Fail	R	Internal power supply failure on Delta DIB.	-	Faulty SMPS. Fault DIB.	
261	Delta 3 DIB +15v Fail	R	Internal power supply failure on Delta DIB.	-	Faulty SMPS. Fault DIB.	
262	Delta 3 DIB -15v Fail	R	Internal power supply failure on Delta DIB.	-	Faulty SMPS. Fault DIB.	
263	Delta 3 DIB Fan Fail	R	Speed feedback failure from capacitor fan for this Delta.	-	Faulty, blocked or jammed capacitor fan. If no fan is present or required, is the jumper link, TP14, on the Delta DIB in the correct position to disable the fan trip detection?	
264	Delta 3 DIB Over Freq. U	R	Erroneous PWM pulses have been detected by the DIB, on this phase, resulting in PWM frequency too high.	-	Incorrect screening of ribbons between controller and DIB. Screen is not connected to metalwork.	
265	Delta 3 DIB Over Freq. V	R	As Trip 244.	-	As Trip 244.	
266	Delta 3 DIB Over Freq. W	R	As Trip 244.	-	As Trip 244.	
267	Delta 3 DIB Td Error U	R	Erroneous PWM pulses have been detected by the DIB, on this phase, resulting upper/lower dead-time violation commands.	-	Incorrect screening of ribbons between controller and DIB. Screen is not connected to metalwork.	
268	Delta 3 DIB Td Error V	R	As Trip 247.	-	As Trip 247.	
269	Delta 3 DIB Td Error W	R	As Trip 247.	-	As Trip 247.	
270	Delta 4 DIB 5v Fail	R	Internal power supply failure on Delta DIB.	-	Faulty SMPS. Fault DIB.	
271	Delta 4 DIB +15v Fail	R	Internal power supply failure on Delta DIB.	-	Faulty SMPS. Fault DIB.	
272	Delta 4 DIB -15v Fail	R	Internal power supply failure on Delta DIB.	-	Faulty SMPS. Fault DIB.	
273	Delta 4 DIB Fan Fail	R	Speed feedback failure from capacitor fan for this Delta.	-	Faulty, blocked or jammed capacitor fan. If no fan is present or required, is the jumper link, TP14, on the Delta DIB in the correct position to disable the fan trip detection?	
274	Delta 4 DIB Over Freq. U	R	Erroneous PWM pulses have been detected by the DIB, on this phase, resulting in PWM frequency too high.	-	Incorrect screening of ribbons between controller and DIB. Screen is not connected to metalwork.	
275	Delta 4 DIB Over Freq. V	R	As Trip 244.	-	As Trip 244.	
276	Delta 4 DIB Over Freq. W	R	As Trip 244.	-	As Trip 244.	
277	Delta 4 DIB Td Error U	R	Erroneous PWM pulses have been detected by the DIB, on this phase, resulting upper/lower dead-time violation commands.	-	Incorrect screening of ribbons between controller and DIB. Screen is not connected to metalwork.	
278	Delta 4 DIB Td Error V	R	As Trip 247.	-	As Trip 247.	
279	Delta 4 DIB Td Error W	R	As Trip 247.	-	As Trip 247.	
280	Delta 5 DIB 5v Fail	R	Internal power supply failure on Delta DIB.	-	Faulty SMPS. Fault DIB.	
281	Delta 5 DIB +15v Fail	R	Internal power supply failure on Delta DIB.	-	Faulty SMPS. Fault DIB.	
282	Delta 5 DIB -15v Fail	R	Internal power supply failure on Delta DIB.	-	Faulty SMPS. Fault DIB.	

Table 9-3. – Trip Fault Codes

Fault Code	Name	Class	Description	Parameter No. to enable Auto-reset	What can cause a trip	Remarks
283	Delta 5 DIB Fan Fail	R	Speed feedback failure from capacitor fan for this Delta.	-	Faulty, blocked or jammed capacitor fan. If no fan is present or required, is the jumper link, TP14, on the Delta DIB in the correct position to disable the fan trip detection?	
284	Delta 5 DIB Over Freq. U	R	Erroneous PWM pulses have been detected by the DIB, on this phase, resulting in PWM frequency too high.	-	Incorrect screening of ribbons between controller and DIB. Screen is not connected to metalwork.	
285	Delta 5 DIB Over Freq. V	R	As Trip 244.	-	As Trip 244.	
286	Delta 5 DIB Over Freq. W	R	As Trip 244.	-	As Trip 244.	
287	Delta 5 DIB Td Error U	R	Erroneous PWM pulses have been detected by the DIB, on this phase, resulting upper/lower dead-time violation commands.	-	Incorrect screening of ribbons between controller and DIB. Screen is not connected to metalwork.	
288	Delta 5 DIB Td Error V	R	As Trip 247.	-	As Trip 247.	
289	Delta 5 DIB Td Error W	R	As Trip 247.	-	As Trip 247.	
290	Delta 6 DIB 5v Fail	R	Internal power supply failure on Delta DIB.	-	Faulty SMPS. Fault DIB.	
291	Delta 6 DIB +15v Fail	R	Internal power supply failure on Delta DIB.	-	Faulty SMPS. Fault DIB.	
292	Delta 6 DIB -15v Fail	R	Internal power supply failure on Delta DIB.	-	Faulty SMPS. Fault DIB.	
293	Delta 6 DIB Fan Fail	R	Speed feedback failure from capacitor fan for this Delta.	-	Faulty, blocked or jammed capacitor fan. If no fan is present or required, is the jumper link, TP14, on the Delta DIB in the correct position to disable the fan trip detection?	
294	Delta 6 DIB Over Freq. U	R	Erroneous PWM pulses have been detected by the DIB, on this phase, resulting in PWM frequency too high.	-	Incorrect screening of ribbons between controller and DIB. Screen is not connected to metalwork.	
295	Delta 6 DIB Over Freq. V	R	As Trip 244.	-	As Trip 244.	
296	Delta 6 DIB Over Freq. W	R	As Trip 244.	-	As Trip 244.	
297	Delta 6 DIB Td Error U	R	Erroneous PWM pulses have been detected by the DIB, on this phase, resulting upper/lower dead-time violation commands.	-	Incorrect screening of ribbons between controller and DIB. Screen is not connected to metalwork.	
298	Delta 6 DIB Td Error V	R	As Trip 247.	-	As Trip 247.	
299	Delta 6 DIB Td Error W	R	As Trip 247.	-	As Trip 247.	

NOTES:

1 Also refer to 6.11.2. Temperature Feedback – Parameters P11.04 to P11.14.

2 A SkiiP® is an integrated assembly of IGBTs.

3 ∗ These faults are hardware generated in the SkiiP® Transistor Module.

4 For any DELTA SkiiP® Trip Fault Code, when the DELTA CDC is operating from 24 V auxiliaries, all DELTA SkiiP®s will report faulty because they are without mains power.

9.6 USING THE HELP KEY

If the drive trips, get information on the trip by pressing ?.

A screen appears, giving four choices as shown in Figure 9-1.

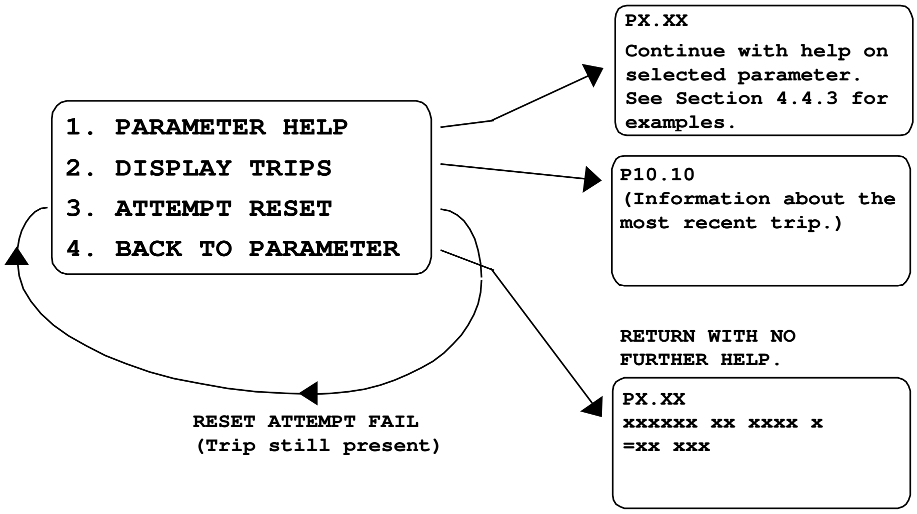


Figure 9-1. – Help Screen for Trips

PRESS

1 Allows the user to view parameter help when a trip is present.

2 Displays parameter P10.10 – gives information about the trip.

3 Attempts to reset the trip. If the attempt fails, this screen re-appears.

4 Returns the parameter being viewed before ? was pressed, with no further help.

If the drive shows a Warning, get information on the warning by pressing ?

The HELP system works as described for trips, except that P10.00 displays the most recent Warning, as in Figure 9-2.

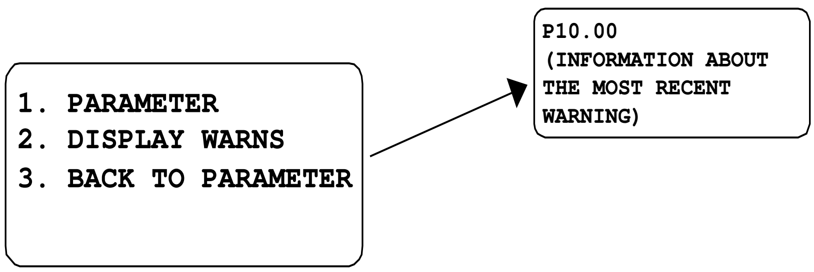


Figure 9-2. – Display of the most recent Warning

9.7 HISTORY LOG

The History Log is a user-defined record of the performance of the drive over a period of time. The method of setting up a History Log is described in Section 6.26.

9.7.1 Playing Back the History Log

To read a sample from the History Log:

a) Set the sample number required in P27.00. (The sample number applies to all ten history channels.)

b) Select the required history channel and read the value. (The meaning of this value will depend on how the History Log is configured.)

c) To read other samples in the selected history channel record, press ↵, then press ▲ or ▼ on the Navigation key to scroll through all the values in the record.

d) To exit this mode, press ESC .

Relevant Parameters are:

P27.00 - History Display Sample Number

Contains the sample that is displayed when the History Log (P27.01 to P27.10) is interrogated.

Allowed values are: 0 to 99 (sample number).

This sample number can be set by selecting and writing to P27.00 in the usual way or it can be incremented and decremented by pressing ▲ and ▼ on the Navigation key while a history record is displayed. This provides a convenient method of scrolling through all the samples in a record.

P27.01 - History Channel 1 Playback Data
P27.02 - History Channel 2 Playback Data
P27.03 - History Channel 3 Playback Data
P27.04 - History Channel 4 Playback Data
P27.05 - History Channel 5 Playback Data
P27.06 - History Channel 6 Playback Data
P27.07 - History Channel 7 Playback Data
P27.08 - History Channel 8 Playback Data
P27.09 - History Channel 9 Playback Data
P27.10 - History Channel 10 Playback Data
P27.11 - Single Element Playback Channel 1 Data
P27.12 - Single Element Playback Channel 2 Data

These parameters each contain up to 100 samples of history data. When a history channel is interrogated, the sample number displayed is determined by the value in P27.00 (for P27.01 to P27.10) and in P26.28, P26.29 for single element channels.

The meaning of the data display for each sample is determined by the configuration defined in P26.01 to P26.20, or in P26.28 and P26.29 for the single element channels.

9.8 DIAGNOSTIC HINTS

Refer to Table 9-4 for helpful hints on fault diagnosis. These hints do vary for each mode of drive operation and so each separate product manual includes a table of diagnostic hints specific to the particular product. However, Table 9-4 includes a composite list of the hints for all product operating modes. The mode is highlighted in the 'Problem' column of the table. The diagnostic hints are aimed mainly at the DEFAULT drive, to help with problems which may be experienced while working with this manual.

Table 9-4. – Diagnostic Hints

Problem	Possible solution
All: Healthy LED is not lit and Tripped LED is lit.	Is the plant "INTERLOCK" connected? Refer to Section 9.4 in this manual and the Wiring Diagram (for Inputs/Outputs) shown in the relevant manual.
All: Drive/SFE failure after frequent switching on & off.	The drive/SFE has correctly carried out self-protection procedures after frequent switching on and off in short period of time – specification is no more than 3 times in any 10 minutes and not more than 10 times in one hour. Replace pre-charge fuses; see 'Spare Parts' in relevant manual.
All except SFE: The drive/SFE will not run from Keypad.	The drive/SFE must be in "Keypad control" (i.e. Local control). Check that DIGIN 4 is open. Use a DVM or view P11.21. DIGIN 4 is connected to CF116 that selects local/remote.
All: The drive/SFE will not run from the terminals.	The drive/SFE must be in "Remote control". Check that DIGIN 4 is closed, use a DVM or view P11.21. DIGIN 4 is connected to CF116 that selects local/remote (or Keypad/remote for SFE). Also for AEM Machine Bridge check that the "SFE Running" output is not inhibiting the "STOP" input on DIGIN 1.
SFE: The SFE will not run from either Keypad or terminals, and displays Warning 136 (CF25/LCN feedback loss)	LCN feedback signal LCN AUX missing: Check line contactor feedback wiring to TB3/6. Check that LCN has closed. If the LCN has not closed: Check LCN coil supply voltage. Check wiring to TB7. Check that DC link volts (P51.00) is greater than pre-charge threshold (P52.17).
SFE: Trip code 96 (Aux phase loss) is displayed and cannot be reset.	SFE is not detecting the correct phase displacement on its auxiliary terminals: Check wiring to AUX R, AUX S, AUX T Terminals. Check pre-charge fuses; see "Spare Parts" in the AEM Getting Started Manual (T2002).
All: All LEDs flashing	This indicates a major software or hardware fault with the controller. Normal software operation cannot continue. Refer to Diagnostics – Failure of Firmware.
All except SFE: The speed reference is not working.	The default drive has 3 speed references programmed: (a) Local (Keypad) reference value entered in P1.00 (b) Remote ANIN1, programmed to be 0 - 10V, view value in P11.36 (c) Remote ANIN2, programmed to be 4 - 20mA, view value in P11.37 Monitor P9.00 whilst operating the required reference. To achieve Keypad reference, ensure DIGIN 4 is open. To achieve any Remote reference, ensure DIGIN 4 is closed. To select between Remote references ANIN1 and ANIN2, toggle DIGIN 5.
All: The analogue input references are not functioning as expected.	The default setting for ANIN1 is 0 - 10 V (0 - 100 %) and the default for ANIN2 is 4 - 20mA (20% - 100 %). The dipswitches (SW1) on the drive I/O board configure the inputs for current or voltage. Check the dipswitches against those shown at the Wiring Diagram (for Inputs/Outputs) shown in the relevant manual. Ensure link TB6/3 to TB6/7 is connected when using the drive's 10.5 V supply (TB6/9). Check the analogue input settings in P7.00 to P7.07 against default settings. Check the values entering the analogue inputs in P7.03, P7.07 respectively.
All: The analogue outputs are not functioning as expected.	The default settings for ANOP1 & ANOP2 are 0 - 10 V, 0 - 100 %. The dipswitches (SW1) on the I/O board should both be set for volts (see Wiring Diagram in relevant manual). Check dipswitches against those shown at Wiring Diagram in relevant manual. Check the analogue output settings in P7.17 to P7.26 against default settings. Check the values coming from the analogue outputs in P7.21, P7.26 respectively.
SFE: SFE trips on over current or over volts when 'run' is attempted.	Check that the Auxiliary terminals (AUX R, AUX S, AUX T) are consistent with the power terminals R/U, S/V and T/W, i.e. AUX R is wired to the same phase of the supply as R/U etc. Check the wiring of ancillary components against the interconnection diagram.
SFE: Trip Code 93 (Mains Sync Fault) is displayed when run is attempted.	Check wiring of ancillary components against the interconnection diagram. Is there excessive impedance in the mains supply to the SFE? (i.e. is the mains supply fault level too small?). See performance data in the AEM Getting Started Manual (T2002).

Table 9-4. – Diagnostic Hints

Problem	Possible solution
All except SFE: Deceleration ramps not being followed, seems to take longer than set.	The drive is programmed at the factory to prevent itself tripping on over voltage trips. When an AC motor is decelerated, the motor generates voltage back to the drive DC Link; the amount of voltage depends on the speed of the deceleration and the load inertia. If the time taken to stop the load is too long: Check the deceleration rates set in P1.23 or P6.02, P6.03 (repeated). A dynamic brake unit may need to be fitted to achieve the required time. Check the value set in P4.12 (P23.05) and ensure that it is sufficient (note that –0.1 kW limit means "unlimited"). If not sufficient adjust it to the dynamic brake resistor surge rating in kW. A small amount of Watts (an amount the drive alone can absorb) set in P4.12 may be enough to solve the problem.
All except SFE: Motor turns slowly and draws excess current when in Vector control with encoder.	This is known as "Wrongly Phased". Check the motor phasing and check the encoder connections. Refer to the Commissioning flowchart, in the relevant manual, which suggests tests that can be made to verify the encoder integrity.
All except SFE: Drive will not complete a CALIBRATION run.	Although the CAL run should be done off load, for small motors with low inertia it may help to keep the coupling on the motor shaft. Also check the accuracy of the basic motor data entered, especially Mag current, it needs to be in the right order of magnitude.
All except SFE: Not enough torque available after a calibration run. VECTOR MODES ONLY.	Check P21.11, this is the measured COLD value of Rr determined by the CAL run. Check P12.15, this is the calculated HOT value of Rr. The value of P12.15 ≅ 1.4 x P12.11. If this is not true simply edit P12.15 = 1.4 x P12.11. Check P12.02, the value of mains voltage, then check the value of P11.49, this is maximum torque the drive has calculated to be available at the motor. Check P8.00 to P8.03. Values greater than P11.49 will be unattainable. Try enabling Auto Temperature Compensation. Set P12.06 = 1. Increase P12.07 to increase the rate of auto-compensation to achieve required torque.
AEM Machine Bridge Warnings 134 and 136 displayed, trips on code 94 when "RUN" is pressed.	Follow the machine bridge Guided Commissioning procedure in the AEM Getting Started Manual (T2002) Section 5B.4 to choose the correct control mode.

9.8.1 Failure of Drive Firmware

Should a fault develop in the drive firmware, normal software operation will stop. If a Keypad is plugged in it will display as below.

Figure 9-3. – Firmware Integrity Failure

When the 'Firmware Integrity Failure' statement is displayed all four LEDs on the drive will flash together. The drive is equipped with a special firmware 'net' that captures the exact state of the drive memory and processor execution state. This data is extremely useful to support and diagnose the reason why the drive firmware crashed.

The memory contents must be uploaded to a file while the drive is displaying a firmware integrity failure, otherwise the captured data will be lost. It will take about 10 minutes to complete the file. The file may be compressed with PkZip/WinZip if required.

To assist **GE Power Conversion** personnel in diagnosing the cause of the software malfunction the memory contents of the drive can be uploaded to a pc, as detailed in Section 9.8.2, and sent to **GE Power Conversion** at the address shown at the end of this manual.

9.8.2 Uploading the Failed Firmware to a PC

The memory contents can be uploaded to a pc running either Windows 3.1 or 3.11 terminal emulation or Windows 95, NT4, 2000 or XP terminal emulation. Use of a pc with any of these terminal emulations is now described.

The hardware required to carry out the file upload is:

- A pc running any of Windows 3.1, 3.11, 95 or NT4 terminal emulation;
- Terminal emulation program (3.1 or 3.11) or Hyper terminal (Windows '95, NT4, 2000 or XP) accessories;
- RS 232 lead e.g. a GDS1009-4001.

Using a PC with Windows 3.1 or 3.11 Terminal Emulation

a) Start Windows Terminal.

b) From ***Settings***, choose ***Communications***.

c) In the dialogue, change the baud rate to ***9600***; change the flow control to ***Xon/Xoff***. All the other items can be left at default (8 data, 1 stop, No parity), choose a communications port at this point.

d) Check that in ***Settings*** - Emulation, it is at its default = VT-100 [ANSI].

e) From ***Transfers***, choose ***Receive text file***.

f) Name the file.

g) Ensure the MV3000e RS232 port is connected to the PC, via the RS232 lead.

h) A capital "G", will start the E000 memory contents upload.

i) The upload takes about 10 minutes.

j) Press ***Stop*** to terminate the text transfer when complete, this saves the file.

k) After the upload, pressing the "." key will re-start the MV3000e firmware.

l) Return the file to **GE Power Conversion** as described in Section 9.8.1.

m) The Terminal connection can be saved as "MV3000e" for next time.

Using a PC with Windows 95, NT4, 2000 or XP Terminal Emulation

a) Start Windows Hyper Terminal.

b) Follow the "***wizard***" as presented.

c) Name the new connection MV3000e, and choose an icon.

d) Choose a comm. port.

e) Set communications parameters to ***9600*** Baud, ***no parity***, ***1 stop***, ***Xon/Xoff*** control.

f) Select ***File*** - ***Properties***, and from the dialogue, choose the ***settings tab***. Set the emulation to VT100.

g) Ensure the MV3000e RS232 port is connected to the pc, via the RS232 lead.

h) From Transfer, choose Capture Text.

i) Name the file.

j) A capital "G", will start the E000 memory contents upload.

k) The upload takes about 10 minutes.

l) Press ***Stop*** to terminate the text capture when complete, this saves the file.

m) After the upload, pressing the "." key will re-start the MV3000e firmware.

n) Return the file to **GE Power Conversion** as described in Section 9.8.1.

NOTE: **If any problems occur when setting the communication with the "Wizard" it may be worthwhile cancelling the "Wizard" and setting the VT100 emulation before the Baud rate etc. produced a working system**

9.9 EXPECTED TRIP CODES WHEN CDC IS POWERED FROM AUXILIARY 28 V INPUT

9.9.1 Introduction

The MicroCubicle™ CDC is a self-contained Printed Circuit Board (PCB), which contains circuits for the controller, the I/O, the power interfaces; connection to Fieldbus boards and an Auxiliary 28V power supply. When powering the MicroCubicle™ CDC from 28V, it is able to determine the size of the drive it is mounted in, by reading vital personality information from an E^{2}PROM mounted on a Switched Mode Power Supply (SMPS) PCB. When the auxiliary 28V is applied, a small part of the SMPS board is also powered up which allows this to happen. The MicroCubicle™ CDC, therefore, knows what type and size of drive it is in, and that affects the trips that it will display as shown in Table 9-5.

Trip code	Meaning	Reason
4	Under voltage	The mains power is off, so the DC link is not established
6	Over-temperature	The SkiiP® circuits are not powered up. Some or all trips may show, depending on the size of the MicroCubicle™ Drive.
9, 11, 13	Hardware Over-temperature U,V,W	
7, 8, 10, 12	Instantaneous Overcurrent, or Instantaneous Overcurrent U,V,W	

Table 9-5. – Expected Trip Codes when Operating a MicroCubicle™ CDC from 28V Auxiliaries

9.9.2 Trip Fault Codes Specific to Internal Dynamic Brakes

Assuming that the Dynamic Brake has been identified and the MicroCubicle™ drive then reverts to 28 V operation after the main AC has been removed, the following trip codes can be expected, depending on the type of internal DB:

- For Frame Sizes 3 and 4 DBs the internal DB will be tripped on trip 151 DB TRIP 26 under temperature.
- For Frame Sizes 6 and 7 DBs the codes in Table 9-6 apply.

Trip code	Meaning	Reason
151	DB trip	The SKiiP® circuits are not powered up. Some or all trips may show, depending on the size of the MicroCubicle™ Drive.
152	DB Hardware Over-temperature	

Table 9-6. – Expected Trip Codes for Internal (to a MicroCubicle™) Dynamic Brakes for Frame Size 6 7 Frame Size 7 Drives

9.9.3 DELTA CDC

The DELTA CDC is a different construction to the MicroCubicle™ CDC. The different functions have been separated to allow the DELTA CDC to be connected to different interfaces for future product enhancements. The CDC and a Power Interface Board (PIB) – common for the whole DELTA range – is housed in a metal container. Any optional Fieldbus boards will also be contained within this container. The I/O has been separated. To run a DELTA CDC on 28 V, apply the 28 V to TB2 on the External I/O. The DELTA CDC determines drive size and type from vital personality information, which is distributed around the DELTA System on different PCBs. (The DELTA CDC can only read the drive size information when powered via the main SMPS).

In summary:

- The DELTA Power Interface Board (DELTA PIB) informs the DELTA CDC that the drive is a DELTA drive. The number of DELTAs is determined by the number of connections made to the ribbon headers on the PIB (a maximum of 6), this information is placed into P99.00.
- The DELTA Interface board (DELTA DIB) is mounted on the DELTA transistor bridge and connects to the IGBT SKiiP® s. This PCB informs the CDC about the current rating of the DELTA, connected to the PIB. The drive nominal current is then reported in P99.05.
- The DELTA Switched Mode Power Supply (DELTA SMPS) informs the DELTA CDC of the voltage rating of the DELTA, and is used to set the Dynamic Brake voltage threshold in P23.04.

When all the information is collected the DELTA CDC has a complete picture of the DELTA system, automatically.

When powered from 28 V, the DELTA CDC can only power up the PIB board, thus it only knows how many DELTAs there are, so P99.00 is correct. The nominal current will report 1000 A, a mid value (not the actual rating of the DELTA). If the DELTA CDC is not connected to any DELTAs, it will still report that 1 DELTA is connected. When programming is completed and the mains power is applied, then the full system configuration can be determined and it will be different from the “assumed” configuration whilst powered from 28 V. Figure 9-4 shows the DELTA Configuration and the state of the User Edits at each stage. Table 9-7 shows the expected Trip Fault Codes when operating the DELTA CDC from 28 V auxiliaries.

In conclusion, any time the controller is powered from 28 V, then re-connected to the mains power, the user edits will be retained, even if the actual DELTA configuration has changed in the meantime.

See Section 10.6 for further information on powering a DELTA Controller from an auxiliary 28V.

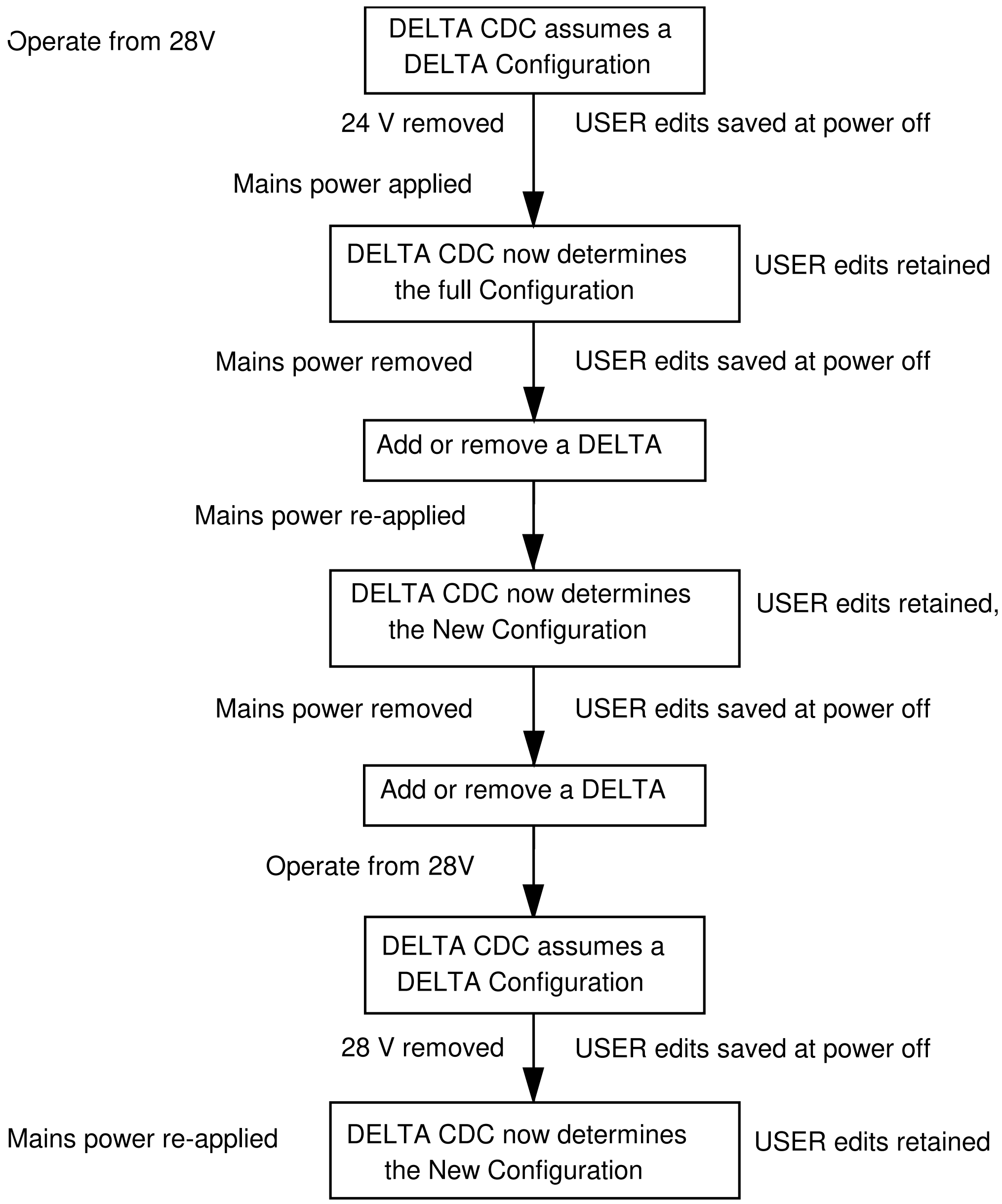


Figure 9-4. – DELTA Configuration showing the state of User Edits

Trip code	Meaning	Reason
4	Under voltage	The mains power is off, so the DC link is not established.
183, 185 …etc	Unidentified SMPS 1, 2, …	The DELTA CDC is unable to read the E^2PROM on the DELTA SMPS when powered from 28 V.
182, 184	Unidentified DELTA 1, 2….	The DELTA CDC is unable to read the E^2PROM on the DELTA DIB when powered from 28 V.
Any DELTA SKiiP® Trip code	All DELTA SKiiP®s will report in faulty	Without mains power the DELTA SKiiP®s will all report in as faulty

Table 9-7. – Expected Trip Fault Codes when operating a DELTA CDC from 28V Auxiliaries

9.10 PRODUCT IDENTITY RECORD

9.10.1 Introduction

The MV3000e product has been designed to minimise the number of parts required to be held for spares. All support staff should be made aware of the following:

a) Drive identity is not held in the drive controller, but distributed around the drive;

b) A serial EEPROM on a printed circuit board contains data relating to the current and voltage rating of the respective MicroCubicle™ or DELTA Module;

c) The EEPROM data is referred to as the 'Product Identity Record';

d) Because of the distribution of the drive identity the CDC and the SMPS must never be removed from the drive together or the Product Identity Record will be lost and will have to be re-entered manually.

9.10.2 Product Identity

The EEPROM includes all the details about the drive in which it is located. It is a collection of 43 items of information stored in the EEPROM during drive/DELTA test when manufactured. The details are unique to the drive or DELTA product and are used by the CDC to set internal drive size dependent parameters, set up feedback scaling and protect the power bridges.

9.10.3 Where the Drive Identity is Stored

In a MicroCubicle™ product

In a MicroCubicle™ product the whole of the Product Identity Record is stored on the SMPS board. The EEPROM circuit is powered from the CDC power supply, so when the 28 V auxiliaries are applied to the CDC, the EEPROM data can be read, so a faulty SMPS is unlikely to prevent the identity from being recovered.

In a DELTA System

A DELTA System has to behave as a complete drive, but DELTA is made up of a number of separate components, and so the identity has to be spread between these components. The DELTA SMPS stores voltage grade information about itself and the DELTA Interface Board (DIB) stores voltage grade information about the DELTA Module and DELTA rating information. All of the DELTAs have SMPS and DIB boards. The 24 V supply cannot power the EEPROMS on the DELTA modules, so mains power has to be applied to a DELTA drive before automatic configuration can take place. The EEPROM data stored in the product is summarised in Table 9-8.

If the Identity of a drive is lost, or invalid, or from a unit of a different type, then the CDC will set itself up incorrectly and the drive will not operate correctly, **this data is therefore vital and must not be corrupted or lost at any time.**

Location of EEPROM	Contents of PCB Specific Record	Product Identity Record
SMPS In MicroCubicle™ or DELTA drives	Calibration data about that specific PCB which removes the need for mechanical potentiometers to scale feedback circuits. PCB serial number and combination number. Voltage grade information.	**In a MicroCubicle™ Drive** SMPS also contains the product identity record, put in during manufacture. **In a DELTA** This part of the EEPROM is left empty during test.
DELTA Interface Board DIB) DELTA drives only	Calibration data about that specific PCB which removes the need for mechanical potentiometers to scale feedback circuits. PCB serial number and combination number.	**In a MicroCubicle™ Drive** There is no DIB fitted in a MicroCubicle™ drive. **In a DELTA** The DIB contains the product identity record, put in during manufacture.

Table 9-8. – Summary of EEPROM Data Stored in MV3000e Products

9.10.4 Why is the identity copied to the CDC

There are two major reasons why the data need to be copied into the CDC:

- So that changes to the drive hardware can be identified;
- So that it can be put back into the SMPS or the DIB if the board is replaced with a 'blank' spare from the factory, or if it is replaced by one removed from another drive.

9.10.5 How is the Drive Identity used and kept safe

When the drive powers up, this information is read by the CDC and is generally used as follows:

a) The calibration data is read and used to scale feedback signals;

b) The voltage grade data sets the Under voltage trip level and Dynamic Brake threshold voltage (P23.04);

c) The Product Identity Record Information allows protection of the drives silicon and allows parameters like P99.05 "Drive Nominal Current" to be set.